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ORIGINAL ARTICLE

Fine-scale spatial genetic structure of white-tailed deer in a national park in the southeastern United States

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Abstract

Large herbivores can become overabundant in sanctuaries, increasing resistance to movement, restricting gene flow among social groups, and leading to spatial genetic structure. We hypothesized that high white-tailed deer (*Odocoileus virginianus*) abundance would cause social resistance to movement, limit gene flow among social groups, and create fine-scale population structure. We tested the prediction that the landscape genetic relatedness of white-tailed deer populations is non-panmictic, characterized by small patches of relatively high genetic relatedness. We collected and geo-referenced 448 fresh fecal samples from white-tailed deer in Vicksburg National Military Park, Vicksburg, Mississippi, USA, in January and March 2023. We genotyped 448 fecal samples with eight polymorphic microsatellite markers, assigning 407 samples to individual deer with unique genotypes. White-tailed deer allelic frequencies were positively correlated within 300 m. Stochastic partial differential equation models predicted a spatially explicit, non-panmictic white-tailed deer population with many small patches (<1 km), where mean genetic relatedness was positive or similar to that of matrilineal groups. High deer abundance may impede movement of individuals in search of resources to peripheral social groups and further restrict gene flow in the Vicksburg National Military Park, where white-tailed deer are protected from recreational hunting and other anthropogenic disturbances.

Keywords Genetic diversity · Landscape genetics · Landscape relatedness · Spatial genetic autocorrelation

Introduction

Genetic diversity is an essential factor influencing mammalian population viability and adaptation to changing environments (Frankham et al. 2002; Freeland 2005; Hughes et al. 2008). Variation in the genetic diversity of wildlife populations is influenced by historical and contemporary demographic events, source populations (or founders), population fluctuations, genetic bottlenecks, and dispersal (Kimura and Weiss 1964; Slatkin 1987; Freeland 2005). Furthermore, restricted dispersal or gene flow may result in non-random spatial variation in genetic diversity across landscapes with important management and conservation implications (Archie and Chiyo 2012; Parreira and Chikhi 2015). For instance, life-history characteristics like mating systems or social behaviors that cause animals to congregate and form groups (e.g., colony or herd) may cause fine-scale population genetic structuring across landscapes (Storz 1999; Armansin et al. 2020). Studying fine-scale population genetic structure, such as spatial autocorrelation of allelic frequencies and spatially explicit genetic relatedness across the landscape (i.e., landscape relatedness or LandRel), is indispensable for understanding wildlife population genetics and variations in genetic diversity (Norman et al. 2017).

Population genetics theory predicts that the distribution of allelic frequencies reaches an equilibrium in a population

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comprised of randomly roaming, randomly mating individuals (i.e., panmictic population) in a homogenous environment (Wright 1931, 1943; Slatkin 1987). Consequently, the spatial distributions of allelic frequencies are expected to be random across the landscape (Norman et al. 2017). On the other hand, violations of any assumptions above (e.g., geographic barrier and non-random mating) may result in the non-random spatial distributions (i.e., spatial structure) of allelic frequencies (Slatkin 1993). For instance, polygynous mating systems and social groups can promote private alleles and increase their frequencies in a location, increasing fine-scale genetic relatedness or autocorrelation of allelic frequencies. As such, this may enhance landscape- or population-level genetic diversity in mammals by maintaining a diverse array of private alleles (Chesser 1991; Double et al. 2005). For instance, excessive, selective culling can disrupt the social structure (e.g., social group) of African elephants (*Loxodonta africana*), resulting in the loss of genetic diversity (Archie and Chiyo 2012).

Reintroduction and stocking have become a practical measure of conserving genetic diversity and restoring endangered or threatened species (Hedrick and Miller 1992). Those restoration measures can increase allelic diversity and genetic heterogeneity with the effects of diverse founders and population size augmentation (Kolbe et al. 2007). For instance, Youngmann et al. (2020) found that the multiple, sequential stockings have increased the genetic diversity and allelic heterogeneity of the elk (*Cervus canadensis*) populations in Kentucky, United States (US). Stocking not only improves the allelic diversity of existing populations immediately, but also prevents the future loss of genetic diversity by increasing the effective population size and reducing the potential effect of genetic drift (Pero et al. 2022). Furthermore, the effects of diverse founders can persist over multiple generations after reintroduction in the elk population, even without geographic barriers (Muller et al. 2018). Therefore, it is plausible that wildlife populations that have received multiple historic stockings and undergo geographic and/or social isolations may become genetically structured.

White-tailed deer (*Odocoileus virginianus* – hereafter, deer) is one of the most popular game animals in the US. Overexploitation caused dramatic deer population declines in the US, including Mississippi, in the early 19th Century (Deyoung et al. 2003). Restoration efforts with reintroduction or stocking have successfully increased deer abundance, estimated at 1.5 million in Mississippi (MDWFP 2024). The historic stockings of deer in Mississippi included seven different sources from Alabama, Louisiana, North Carolina, Kentucky, Wisconsin, Mississippi, and Mexico (Deyoung et al. 2003). Recreational deer hunting takes place in most public and private lands across Mississippi except for national parks (Demarais et al. 2012). Most national parks have become a sanctuary from recreational hunting and other anthropogenetic disturbances, which often results in high population numbers and even overabundance (Demarais et al. 2012). However, it is unknown whether high abundances and diverse founders from sequential historic stockings have reduced the fine-scale genetic relatedness and genetic structuring of deer in the Mississippi national parks.

Several studies have investigated the fine-scale population genetic structure of deer on managed land (Scribner et al. 1997; Comer et al. 2005; Miller et al. 2010). Those studies have primarily employed spatial autocorrelation analysis to test for fine-scale population genetic structuring of deer (Comer et al. 2005; Miller et al. 2010). Spatial autocorrelation analysis estimates the mean spatial autocorrelation of allelic frequencies, instead of genetic relatedness, within a series of distance classes (Peakall and Smouse 2006). However, spatial autocorrelation analysis does not provide spatially explicit estimates of genetic relatedness (i.e., relatedness at specific georeferenced locations). Such spatially explicit information on fine-scale genetic structure, incorporating explicit landscape information (such as geographic location, size, and shape of spatial genetic clusters), can provide insight into contemporary dispersal or gene flow across the landscape to inform localized management and conservation of wildlife populations (Norman et al. 2017; Rutten et al. 2019; Medinas et al. 2023).

In this study, we investigated both the spatially explicit, fine-scale landscape genetic relatedness (i.e., LandRel) and the spatial genetic autocorrelation of white-tailed deer in the Vicksburg National Military Park in the southeastern US in January and March 2023. We hypothesized that the historic stockings and high deer abundances have created multiple population genetic clusters or divisions of white-tailed deer in the Vicksburg National Military Park (Budd et al. 2018; Chafin et al. 2021). In polygynous mammals, females are more likely to remain and breed in their natal group than males, leading to male-biased dispersal (Clutton-Brock 1989). Females often have their first conception at an older age than the average length of residency for adult males in breeding groups for inbreeding avoidance. Dispersing roe deer (*Capreolus capreolus*) males join the destination groups of unrelated deer for inbreeding avoidance (Biosa et al. 2015). However, Gaillard et al. (2008) did not find evidence supporting sex-biased dispersal in polygynous roe deer based on long-term monitoring data. We hypothesized that social groups result in fine-scale (< 1 km) landscape relatedness and fine-scale spatial genetic autocorrelation through restricted gene flow (i.e., social resistance) between the groups (Sugg et al. 1996; Armansin et al. 2020). Vicksburg National Military Park borders the Yazoo River. Winter rainfall increases the river's water levels and can cause flooding in the surrounding area. Higher water levels and occasional flooding push deer into the park, blending the genetic makeup of white-tailed deer within the park. We used landscape genetic approaches and spatial statistical models to estimate spatially explicit landscape relatedness across the entire landscape, providing insight into white-tailed deer population management (Norman et al. 2017). For instance, intentional removal of social groups creates a persistent white-tailed deer population of low density (McNulty et al. 1997). Such targeted removal can be informed by prior landscape relatedness analysis.

Methods and methods

Study site

Vicksburg National Military Park is located in Vicksburg, Warren County, MS, US (32°20’23” N, -90°51’3” W; Fig. 1). The park is approximately 730 ha in size and is adjacent to human residences to the northeast of Vicksburg. The climate is subtropical with mean annual temperatures ranging from 12 °C to 25.2 °C and annual total precipitation from 73.33 cm to 210.49 cm. The vegetation in the study area is primarily comprised of mesic upland deciduous (or hardwood) forests and grasslands (or open field, Fig. 1). Vicksburg National Military Park is situated on erodible loess bluff lands in the East Gulf Coastal Plain. These adjacent lands are at the southern edge of the Mississippi Alluvial Valley (hereafter, the Mississippi Delta), which is bounded between the Mississippi and Yazoo Rivers. The Mississippi Delta is subject to periodic flooding, particularly during the winter and spring seasons.

Deer fecal sample collection

Vicksburg National Military Park has a high relief terrain with elevation ranging from 18 m to 150 m a.s.l. (mean = 96.61, standard deviation [SD] = 20.55; Fig. S1). We divided the study area into four sections—South (S), Middle 1 (M1), Middle 2 (M2), and North (N, Fig. 1). We pre-selected 11, 1–2 mile transects for white-tailed deer fecal sample collection: two transects in each of the N, M1, M2, and S sections, parallel to the paved tour roads, and

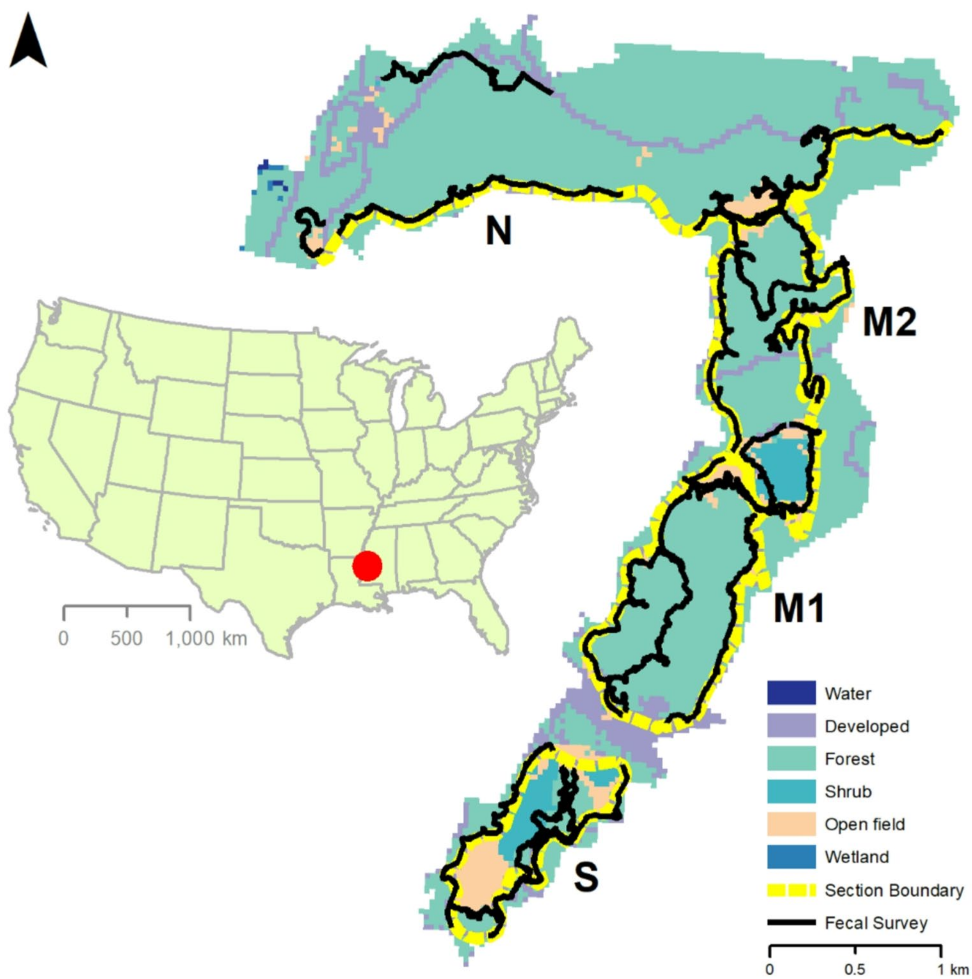

Fig. 1 White-tailed deer fecal sample search routes in the Vicksburg National Military Park, Mississippi, USA, in January and March 2023. Yellow dashed-line polygons or lines delineate the divisions of four sections of the park. Black solid lines are the fecal sample search transects. The land use and land cover map was derived from the National Land Cover Database 2021. The arrow at the upper left top indicates the direction of the North. Symbols or letters M1, M2, N, and S stand for the Middle 1, Middle 2, North, and South sections of the park, respectively. The inset of the United States map indicates the geographic location of the park

one transect along the Al Scheller Trail (colloquially called the Boy Scout Trail) in each of the N, M1, and M2 sections to cover the interior parts of the landscape of the study area (Figs. 1 and S1). Due to the rough terrain of Vicksburg National Military Park, we were not able to randomly place fecal sample search transects within the study area (Fig. S1). Two observers walked each transect to search for deer fecal pellet piles along the tour road, off the tour road, and along the main trail, all of which present various vegetative regimes, particularly manicured cool- and warm-season native and non-native grasses, and forested areas. The observers followed white-tailed deer tracks or trails until they found fecal piles. We considered fecal pellets scattered within 20 cm as a fecal sample from the same deer. We collected wet, shiny, intact pellets with a sheen (mucus coating layer) and discarded dry, hard, eroded, or cracked pellets. We collected ten pellets from each pile and stored the pellets of each detected pile in a 50-mL sterilized plastic centrifuge tube with a screw-on cap. We marked each tube with a unique sample identification (ID) number, longitude and latitude of pellet sample location, transect ID number, observer initials, and date with a fine-tip Sharpie marker. We also recorded the sample information of all collected fecal samples on the data sheets separately. Observers wore a new pair of fresh powder-free nitrile examination gloves to collect each fecal pile to avoid cross-contamination. We searched each of the 11 fecal sample transects twice for fecal samples during January 5–10, March 10–19, and May 10–19 of 2023. Fecal samples were stored in a refrigerator within two hours and were transported to Mississippi State University, where they were stored in a -20 °C freezer within three days after collection.

Extraction and microsatellite analysis of fecal DNA for genotyping white-tailed deer fecal samples and deer individual assignments

DNA extraction

We extracted DNA from the epithelial cells of six pellets from each sample using QIAamp Fast DNA Stool Mini Kit (Qiagen, Valencia, CA). DNA extraction followed the manufacturer's protocol. Before conducting PCRs for genotype analysis, we checked the DNA's quantity and quality with the Nanodrop instrument, performed PCRs for each chosen microsatellite marker (described in the next section), and then compared the band patterns to those from the muscle samples of deer harvested from the same region using electrophoresis with an agarose gel in the laboratory at Mississippi State University. Extracted DNA was stored in a -20 °C freezer for the subsequent PCR runs of microsatellite markers.

Genotyping and microsatellite analysis

We amplified 11 microsatellite markers (BM4107, RT5, SBTD04-A565, BL25, CERVID1-A550, INRA011, BL42-A565, BM848, SBTD06, BM6506, and BM6438) from fecal DNA to genotype and identify individual deer (Bishop et al. 1994; Goode et al. 2014; Miller et al. 2019). We amplified the 11 microsatellite markers using a published PCR protocol in the described cycling and buffer conditions (Bishop et al. 1994; Goode et al. 2014; Miller et al. 2019). The genotype analysis of the 11 microsatellite markers was conducted using the ABI Genetic Analyzer 3730 (Applied Biosystems, Grand Island, NY, USA) with an internal size standard (LIZ-500, Applied Biosystems) at the Center for Quantitative Life Sciences at Oregon State University (Corvallis, OR). We used the program *GeneMarker* ver. 2.6.7 (Softgenetics Inc., State College, PA) to score the allele sizes of each microsatellite locus to determine the genotype of each fecal sample. We replicated PCR amplification for each fecal DNA sample and ran additional PCRs on the DNA samples, which had inconsistent results in the first two PCRs following Maudet et al. (2004). We repeated PCRs three times for homozygous microsatellite loci. We assessed the probability for genotyping errors, using the software *Micro-Checker* ver. 2.2 (Van Oosterhout et al. 2004).

Unique multi-locus genotype identifications and individual assignments of deer fecal samples

Several of the following analyses were conducted in the R 4.4.1 environment (R Development Core Team 2024). We used the R package *allelematch* ver. 2.5.2 to identify unique multi-locus genotypes (i.e., genotyped individual deer) among the deer fecal samples (Galpern et al. 2012). We calculated the probability of identity of individuals and the probability of identity for siblings with *allelematch*. If the probability of identity is $< 10^{-7}$ and the probability of identity for siblings is $< 10^{-3}$, we concluded the individual assignment of deer fecal samples was reliable (Kalinowski et al. 2006). We also used the R package *poppr* ver. 2.9.6 to determine the minimum number of microsatellite loci that was sufficient to identify unique multi-locus genotypes of the fecal samples (Kamvar et al. 2015). We used the program *GenAlEx* v.6.4 added on to Microsoft® Excel to calculate the number of unique alleles, the number of effective alleles, observed heterozygosity, expected heterozygosity, and Shannon's index of allelic diversity by microsatellite locus (Peakall et al. 2003; Peakall and Smouse 2006).

Population genetic clustering

We determined the number of population genetic divisions of Vicksburg National Military Park deer using the discriminant analysis of principal components (DAPC), available in the R package *adegenet* ver. 2.1.10 (Jombart et al. 2010). The DAPC method is a multivariate statistical method for population genetic clustering based on principal component analysis, which does not require Hardy-Weinberg equilibrium and linkage equilibrium (Jombart et al. 2010). We chose the number of principal components (PCs) that accounted for >90% of genetic variability. We used the Bayesian information criterion (BIC) to determine the optimal number of genetic subdivisions, at which the BIC was the lowest (Jombart et al. 2010). We also retrieved the cluster memberships of genotyped deer individuals from the function dapc() of the R package *adegenet*.

Spatial autocorrelation analysis

We used the geographic coordinates (i.e., longitudes and latitudes) of fecal sample locations as the coordinates of the genotyped white-tailed deer in spatial autocorrelation analysis (Smouse and Peakall 1999; Double et al. 2005). Because a genotyped deer may be detected more than once within the study area during our study, we used the first sampling location for each genotyped deer in spatial autocorrelation and subsequent spatially explicit analyses. We used the program *GenAlEx* to compute the spatial autocorrelation of all unique genotyped deer from 0 to 5,000 m at a distance class size of 100 m (Peakall et al. 2003; Peakall and Smouse 2006). We bootstrapped the 95% confidence interval (CI) of the autocorrelation coefficient for each distance class with 1,000 repetitions. The bootstrap algorithm draws 1,000 random samples with replacement from the subset of the allelic frequencies within each distance class, and estimates the empirical 95% CI (i.e., the 25th and 975th ranked bootstrapped values) for the observed autocorrelation coefficient.

We used the permutation method to test for the significance of spatial autocorrelation with 1,000 repetitions. The permutation algorithm randomly shuffles all individuals among the geographic locations 1,000 times. The permutation test generates the empirical 95% CI (i.e., the 25th and 975th ranked permutated values) of the autocorrelation coefficient under the null hypothesis of no spatial structure or complete spatial randomness. If the bootstrapped 95% CI of the observed autocorrelation coefficient exceeded or was below the permutated 95% CIs under the null hypothesis within a distance class, we concluded the deer had significantly positive or negative spatial autocorrelation within the distance class.

Spatial statistical models for the fine-scale, spatially explicit landscape relatedness of Vicksburg National Military Park deer

Previous studies have demonstrated that the Lynch-Ritland coefficient *r* outperforms other coefficients for empirical data (Thomas 2005; Norman et al. 2017). We calculated pairwise genetic relatedness of white-tailed deer using the Lynch-Ritland coefficient *r* with the R package *related* ver. 1.0 (Lynch and Ritland 1999; Pew et al. 2015). Our fecal sample search transects were not randomly placed over the study site (i.e., preferential sampling). The stochastic partial differential equation (SPDE) models can overcome the shortcomings of the spatial statistical inferences from a preferential sampling with spatially non-random transects (Zhong et al. 2025). Therefore, we used the SPDE models to interpolate the genetic relatedness of white-tailed deer across the landscape within the study area.

The SPDE model uses the Voronoi tessellation mesh to discretize the continuously georeferenced space (e.g., with longitudes and latitudes) to represent the continuous spatial process with a sparse precision matrix (i.e., the inverse of the spatial covariance matrix) (Lindgren and Rue 2015). Furthermore, the SPDE models approximate observed responses at the sample locations or interpolate unobserved responses for non-sample locations using the finite element method (FEM). We fit the SPDE models to the georeferenced genetic relatedness coefficients of white-tailed deer with the R package *R-INLA* ver. 24.06.27 (Integrated Nested Laplace Approximation) (Rue et al. 2009). The INLA is a Bayesian approximation method that avoids the difficulties in the convergence of Markov chain Monte Carlo (MCMC) methods for estimating Bayesian statistics (Rue et al. 2009). The further details of the INLA and SPDE models can be found in Rue et al. (2017) and Lindgren and Rue (2015), respectively.

The SPDE-based spatial statistical methods for the spatial interpolation of the fine-scale landscape relatedness of white-tailed deer within the study area included the following two steps. First, we calculated the Lynch-Ritland *r* between the focal deer (i.e., a uniquely genotyped fecal sample) and each of the remaining known genotyped deer (i.e., *N*-1 *r* values) in our samples across the landscape. We calculated the mean Lynch-Ritland *r* over the *N*-1 pairwise genetic relatedness coefficients for each focal deer (Norman et al. 2017). We repeated the same process for our study's known genotyped white-tailed deer (i.e., *N* times). Second, we fit the SPDE model with the intercept only and Gaussian distributions to the mean Lynch-Ritland *r* values with the 2-D Voronoi tessellation mesh created from the geographic coordinates of the *N* fecal samples, which were reprojected to the northing (m) and easting (m) coordinates of the equal-distance Albers Projection) (Norman et al. 2017). We visualized the spatial distribution of the genetic relatedness of white-tailed deer using a raster image (or a heatmap of landscape relatedness) predicted by the SPDE model.

Spotlight survey for the relative deer abundance

We conducted five to seven spotlight surveys across seven pre-selected survey transects in January, March, and May 2023. Because plant leaf-out occurred throughout April and May, it was difficult to identify deer during the surveys. We only reported results from the January and March surveys. These included a transect along the paved road in the North section (7 km), as well as on the west and east sides of the two Middle sections (M 1: east = 2.4 km; west = 3.4 km; M 2: east = 6.4 km, west = 2.4 km), and the South section (east = 2.7 km, west = 2.1 km). The distances between transects and rugged terrain differences prevent observers from double-counting the same deer. Due to the rugged terrain and tortuous roads, we could not randomly place equal distance transects within the park. Surveys were conducted at night, starting 30–60 min after sunset and ending once all transects were surveyed each night. Each survey team had at least three people: one driver and two observers. Observers stood in the bed of a pickup truck moving at 5–7 miles per hour. Surveys followed the current one-way traffic flow through the park, which was counterclockwise. Each observer used a hand-held Maximum Million III spotlight (Brinkmann Corporation, Dallas, TX) to detect and count deer on one side of the road. To reduce observer bias, each observer was rotated among the four sections during each survey night. We also randomly rotated the starting transect and the order of transects across the 5 to 7 survey nights each month to account for daily changes in deer activity and effects related to survey order.

Deer count per km was calculated as the total number of deer detected on both sides of a transect divided by the actual total length of the transect recorded during each survey. We used deer count per km as a relative abundance measure for the entire park. We calculated the daily deer count per km by transect, along with the monthly averages and standard deviations (SD) over all transects for the entire park and all survey days within each month.

Table 1 Allelic polymorphism and heterozygosity of eight microsatellite markers from white-tailed deer samples in Vicksburg National Military Park, Vicksburg, Mississippi, collected during January and March 2023

Locus name	*N*	Na	I	Ho	He
BL25	399	15	1.93	0.71	0.78
CERVID1-A550	402	25	2.55	0.88	0.90
INRA011	399	11	0.94	0.40	0.44
BM4107	388	21	2.68	0.62	0.91
RT5	407	19	2.30	0.90	0.87
SBTD06	390	11	1.51	0.85	0.73
BM6506	406	23	2.56	0.77	0.89
BM6438	368	19	2.37	0.82	0.86

N: sample size; I: Shannon's diversity index; Na: number of alleles; Ne: number of effective alleles; Ho: observed heterozygosity; He: expected heterozygosity

Results

We collected 507 fecal samples in January ($n = 309$) and March ($n = 198$) in 2023. We discarded 39 fecal samples collected under warm, humid weather in May 2023 because of low quality and the failure of DNA extraction. Of those 507 samples in January and March, 448 fecal samples rendered high-quality DNA for the PCR runs. Of the 11 microsatellite markers, RT5, SBTD04-A565, and BL42-A565 were not successfully amplified in most of the 448 fecal samples, which were widely scattered within the study area (Fig. S2). Subsequently, we used the eight remaining microsatellite markers to genotype 448 fecal samples to identify individual deer (Table 1). The observed and expected allelic heterozygosity ranged from 0.62 to 0.9, except for INRA011 (Table 1), indicating high genetic diversity.

Individual assignment analysis with the R package *allelematch* identified 408 unique genotypes representing 408 deer. The genotype accumulation curve indicated that ≥four microsatellite markers can identify 408 unique genotypes, that is, 408 unique individual deer (Fig. S3). The probabilities of identity and the probabilities of identity for siblings were 1.46×10^{-11} and 0.0003, respectively. Therefore, the individual assignments had negligible errors. Out of the 408 samples, we used 407 genotyped white-tailed deer in the subsequent analysis, excluding one sample due to uncertainty of its geographic coordinates.

Discriminant analysis of principal components identified six clusters having the lowest BIC for the 407 genotyped white-tailed deer (Figs S4-S5). The first and second discriminant components explained 98.4% of the total genetic variability. Spatial autocorrelation of allelic frequencies was significantly positive up to a distance class of 300 m with the bootstrapped 95% CI of mean autocorrelations for 0–100 m and 200–300 distance classes being above those of the null hypothesis (Fig. 2a). Mean pairwise Lynch-Ritland genetic coefficients of all genotyped fecal samples ranged from −0.05 to 0.19 (Fig. 2b). The SPDE model predicted a non-panmictic white-tailed deer population within the study area. Each park section had multiple small patches (< 1 km from east to west or from north to south), in which mean genetic relatedness ranged from 0.05 to 0.1 (Fig. 3). The average number of deer per km was 8.52 (SD = 6.06) and 8.65 (SD = 4.75), respectively, for January and March. The

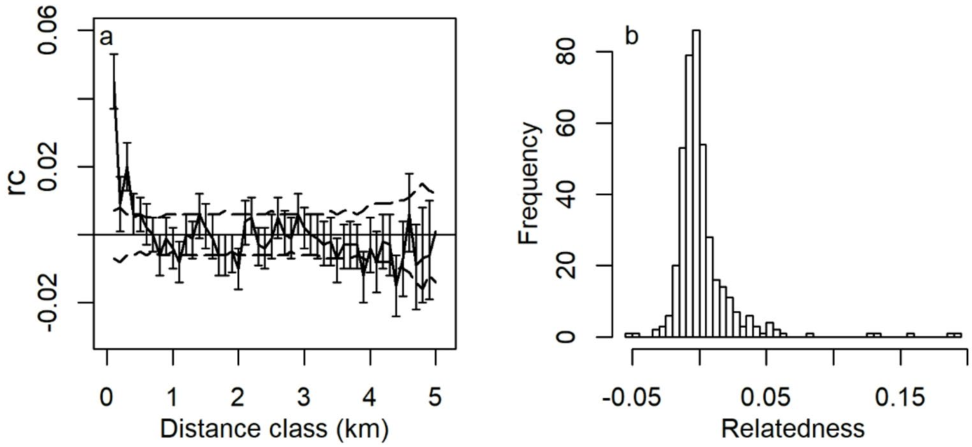

Fig. 2 (**a**) Spatial autocorrelation of the allelic frequencies of white-tailed deer in the Vicksburg National Military Park, Mississippi, USA. The abscissa is the end point of the distance class (100 in length). The ordinate is the spatial autocorrelation coefficient (*rc*) of the gene frequencies of eight microsatellite loci. Two dashed lines delimit the permutated 95% confidence interval band of spatial autocorrelations under the null hypothesis that genotypes are randomly distributed over space. Vertical lines are the bootstrapped 95% confidence intervals of mean spatial autocorrelations. (**b**) Histogram of pairwise Lynch-Ritland genetic relatedness coefficients of genotyped wild-tailed deer in the Vicksburg National Military Park, Mississippi, USA

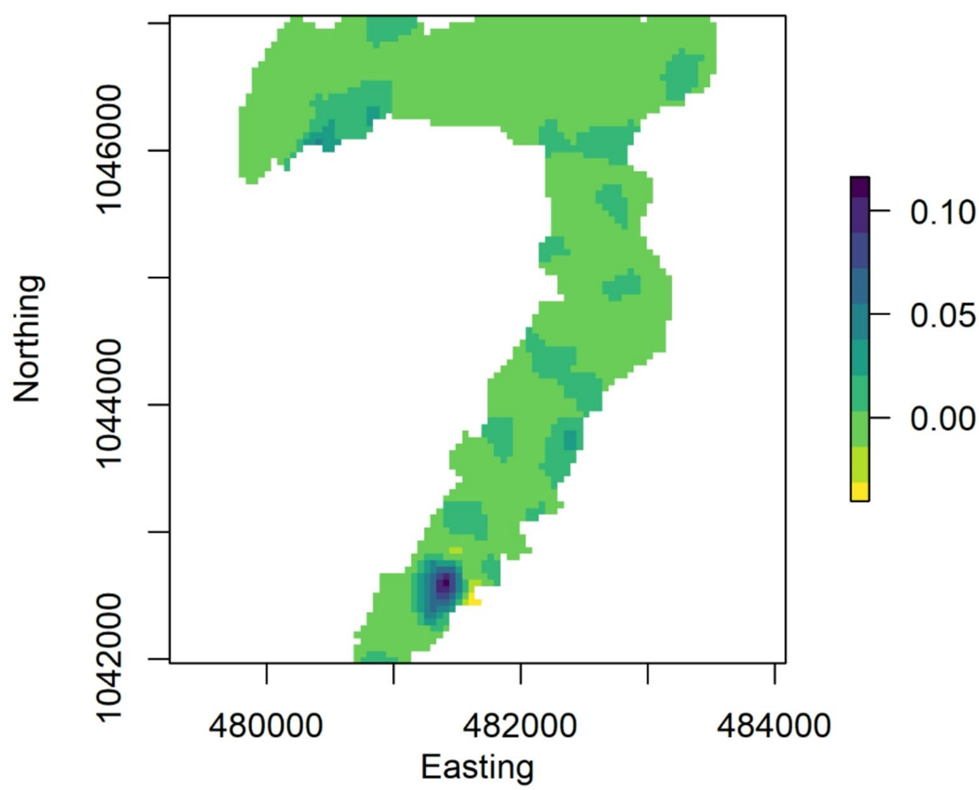

Fig. 3 Landscape genetic relatedness (mean pairwise Lynch-Ritland coefficient) of white-tailed deer in the Vicksburg National Military Park, Mississippi, USA. The Landscape genetic relatedness was estimated and interpolated with the eight-loci microsatellite genotypes of white-tailed deer fecal samples using stochastic partial differential equation models. The legend on the right indicates the mean Lynch-Ritland relatedness coefficient. The abscissa and ordinates of the northing (m) and easting (m) coordinates of the Albers Equal-distance Projection, respectively

relative deer abundance did not differ between January and March ($z=0.16$, $p=0.87$).

Discussion

Knowledge of the spatially explicit, fine-scale genetic structure of wildlife populations helps wildlife ecologists better understand the population genetic consequences of wildlife movements, social groups, abundance, and management actions (Sugg et al. 1996; Double et al. 2005; Norman et al. 2017; Armansin et al. 2020). We revealed a "cryptic," fine-scale (< 1 km in distance) population genetic structure of deer in the study area in the southeastern US using spatially explicit methods. Our findings support our prediction that the deer population in the study area consists of multiple population genetic clusters, contributing to high genetic diversity across the entire park. Our results also support the hypothesis that restricted dispersal or gene flow between social groups creates small (< 1 km) spatial clusters of positive or similar to genetic relatedness across the landscape. These findings suggest that social resistance from social groups, particularly matrilineal groups, may maintain fine-scale genetic segregation, resulting in high genetic diversity like that in African elephants and other group-living mammals (Archie and Chiyo 2012; Parreira and Chikhi 2015; Wang et al. 2023).

The average observed heterozygosity over the eight microsatellite markers in the Vicksburg National Military Park is comparable to that of deer populations in Georgia (Comer et al. 2005), Missouri (Budd et al. 2018), Ohio (Bauder et al. 2021), Tennessee (Goode et al. 2014), and West Virginia (Miller et al. 2010), US, falling within a heterozygosity range of 0.70–0.77. The high heterozygosity of deer may result from historic stockings and high deer abundance in the eastern US. High abundances or population sizes prevent the quick loss of genetic diversity resulting from genetic drift. Social groups also help group-living mammals, including the matrilineal groups of deer, maintain genetic diversity through restricted gene flow and retaining diverse private alleles (Chesser et al. 1993; Sugg et al. 1996; Parreira and Chikhi 2015).

Contemporary population genetic divisions may result from multiple historic stockings, geographic barriers, and social resistance. Multiple lines of evidence have been found that long-distance translocations, like those in Mississippi in the 1960s, have profoundly impacted the genetics of existing deer populations in the southeastern US (Leberg and Ellsworth 1999). Chafin et al. (2021) found that historic translocation was a primary driver of deer population genetic structure in Arkansas, US. Additionally, translocations leave persistent matrilineal genetic signals on the white-tailed deer population genetic structure in Missouri, and probably have resulted in three different genetic clusters in several regions within Missouri (Budd et al. 2018). Unlike those previous studies of population genetic clustering among different areas of a state, our study has revealed population genetic clustering in a local deer population. Vicksburg National Military Park is near one of Mississippi's largest historic deer stocking sites (Deyoung et al. 2003). Periodic flooding of the Yazoo River can cause deer to move into Vicksburg National Military Park during winter and early spring, often mixing different genetic sources of deer within the park. Furthermore, three deer subspecies are distributed within Mississippi; Vicksburg National Military Park is near the boundary of two deer subspecies *O. virginianus virginianus* and *O. v. mcilhennyi* (Baker 1984; Deyoung et al. 2003). Although we did not determine deer's phylogenetic origins and stocking sources in this study, historic stockings along with potentially > 1 subspecies may differentiate the genetic makeup of the Vicksburg National Military Park deer populations. The Wahlund effect and social resistance to gene flow might result in the observed deer population genetic structure, in which the Hardy-Weinberg equilibrium assumption was not met. Lack of information on the phylogenetic origins and stocking sources can cause uncertainties about the effects of historic restocking efforts. Therefore, future research using mitochondrial DNA markers or SNP markers to determine phylogenetic origin and stocking sources is necessary for a clearer understanding of the genetic impacts of past restocking.

Polygynous, group-living mammals may have social resistance to between-group dispersal, resulting in fine-scale spatial genetic structure (Chesser 1991; Sugg et al. 1996; Storz 1999; Armansin et al. 2020). Inbreeding depression has been found in ungulates such as red deer (*Cervus elaphus*) and roe deer (Slate et al. 2000; Da Silva et al. 2009). Inbreeding avoidance can limit gene flow between social groups, especially across a landscape where closely related animals are present, leading to a patchy genetic structure (Biosa et al. 2015). White-tailed deer populations are known to be non-panmictic, exhibiting fine-scale genetic structure in the Savannah River area, Georgia (Scribner et al. 1997) and West Virginia (Miller et al. 2010). The average genetic relatedness of deer fawn-doe groups is about 0.10 in West Virginia, approximating the expected relatedness of first cousins (Miller et al. 2010); consequently, deer populations have a significant positive genetic correlation within a 500-m distance (Miller et al. 2010). The spatial genetic autocorrelation of deer in our study area is similar to that (300 m distance class < 1000 m) of white-tailed deer populations in West Virginia and Georgia, as well as roe deer (*Capreolus capreolus*) and red deer (*Cervus elaphus*) in Europe despite the high vagility of large herbivores (Nussey et al.

2005; Bonnot et al. 2010). Although landscape resistance is a critical factor shaping the spatial genetic structure of wildlife populations, the spatial genetic structure of white-tailed deer is not related to landscape structure in Ohio (Bauder et al. 2021). Therefore, resistance from social groups may be a primary factor attenuating between-group dispersal across the landscape, resulting in fine-scale spatial genetic structuring of white-tailed deer populations.

The relative abundance of deer is three or more times greater than those in the national forests within Mississippi, which were surveyed using the same spotlight methods (Jones et al. 2020). In a survey conducted by the Mississippi Department of Wildlife, Fisheries and Parks in April 2023 using infrared cameras, the relative abundance of deer in the southern 160-km segment of the Natchez Trace Parkway—which is a US national park along a 710-km highway across Mississippi—was 7.3 deer per km (Kamen Campbell, personal communication). National parks have served as refuges or shelters for wildlife, often with overabundant deer (Demarais et al. 2012). Thus, our findings can provide insights into the management of deer and other wildlife in the southeastern US. Our study estimated the genetic relatedness of overabundant white-tailed deer across the landscape in a spatially explicit manner, providing further insights into the effects of social groups on the spatial genetic structuring of deer beyond those inferred by spatial autocorrelation analysis (Norman et al. 2017). The spatial clusters of deer had average genetic relatedness similar to that of deer matrilineal groups (Miller et al. 2010). Therefore, the spatial clusters of high genetic relatedness correspond to habitat patches occupied by deer social groups. The heatmap of landscape relatedness also illustrates the potential social barrier to contemporary gene flow or dispersal, that is, the space or matrix of low genetic relatedness between the spatial clusters. Furthermore, it is plausible to hypothesize that genetic relatedness within immigrating deer matrilineal groups and social resistance caused by high population density may divide the genetic structure of deer populations as observed in this study. Future studies are warranted to test this hypothesis using data on the sex and social group of deer. In conclusion, the spatially explicit landscape relatedness analysis provides further evidence for the primary role of social groups in structuring the fine-scale population genetics of white-tailed deer. The SPDE-landscape ecology approach is a valuable addition to the landscape genetic toolbox for investigations of wildlife dispersal and design of localized wildlife population management (Cayuela et al. 2018; Rutten et al. 2019; Medinas et al. 2023).

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Author contributions XS, GW, and SDB conceived the idea. XS, GW, and WS conducted the study. XS and GW analyzed data. XS drafted the manuscript. XS, GW, SDB, WS, CVB, and SD edited the manuscript.

Data availability The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Animal ethics No live animals were used in this study.

Conflicts of interest The authors have no conflict of interest.

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